Machine Learning Street Talk

The MLP Assumption, Inductive Biases, Self-organizing/Coordinating Networks

Presented By

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The MLP Assumption

MLP

$$f(x) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(w_i.x + b_i)$$

- Linear weights on edges
- Non-linear activation on nodes

KAN

$$f(x) \approx \sum_{q=1}^{2n+1} \Phi_q(\sum p = 1^n \phi_{q,p}(x_p))$$

- Learnable activation functions on edges
- Nodes perform simple summation without non-linearitues

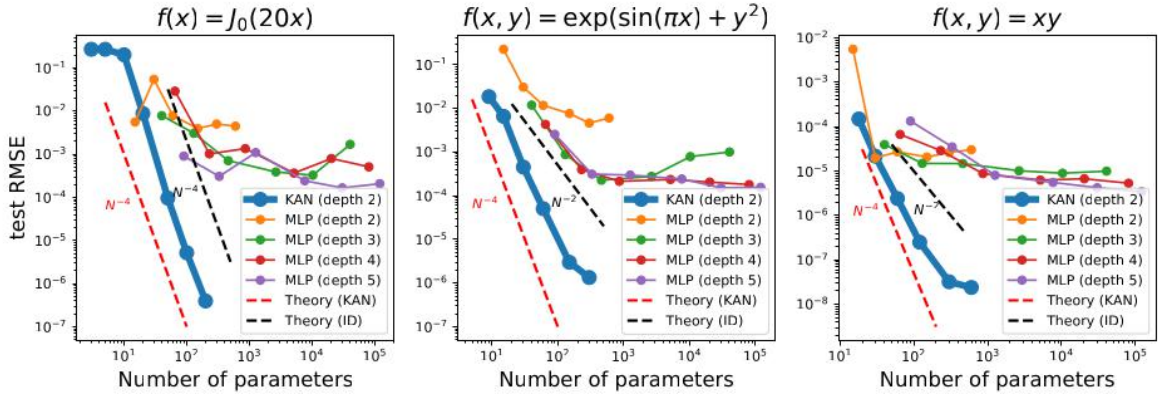
COMPOSITIONAL STRUCTURE

- Hierarchical 1D Functions
- Arbitrary d-dimensional Functions

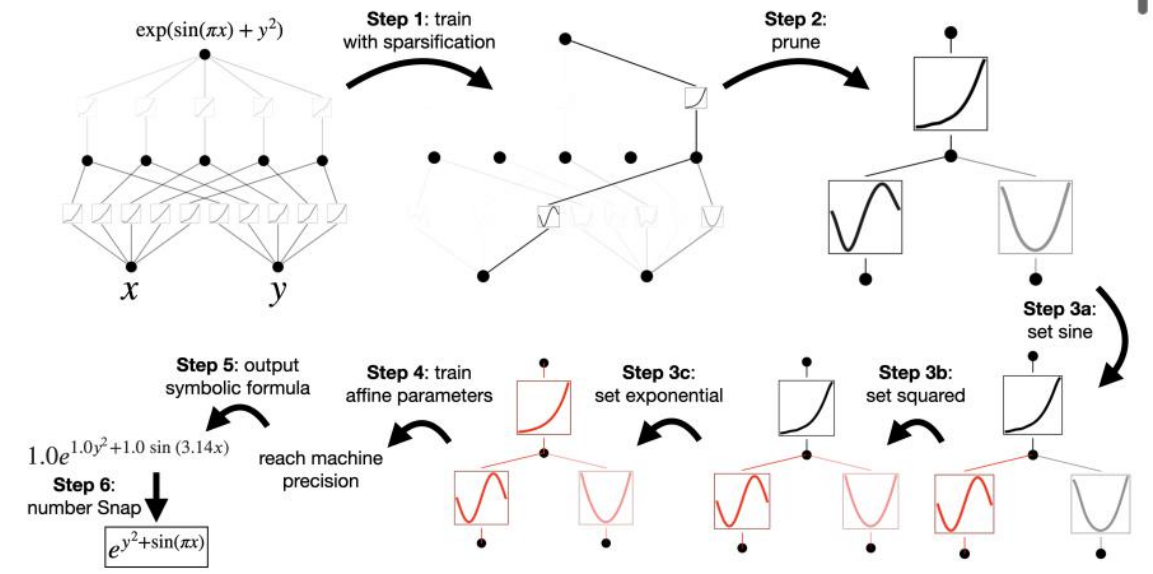
LOCAL PLASTICITY

- Localized B-spline updates
- Global Activations

Parameter Efficiency

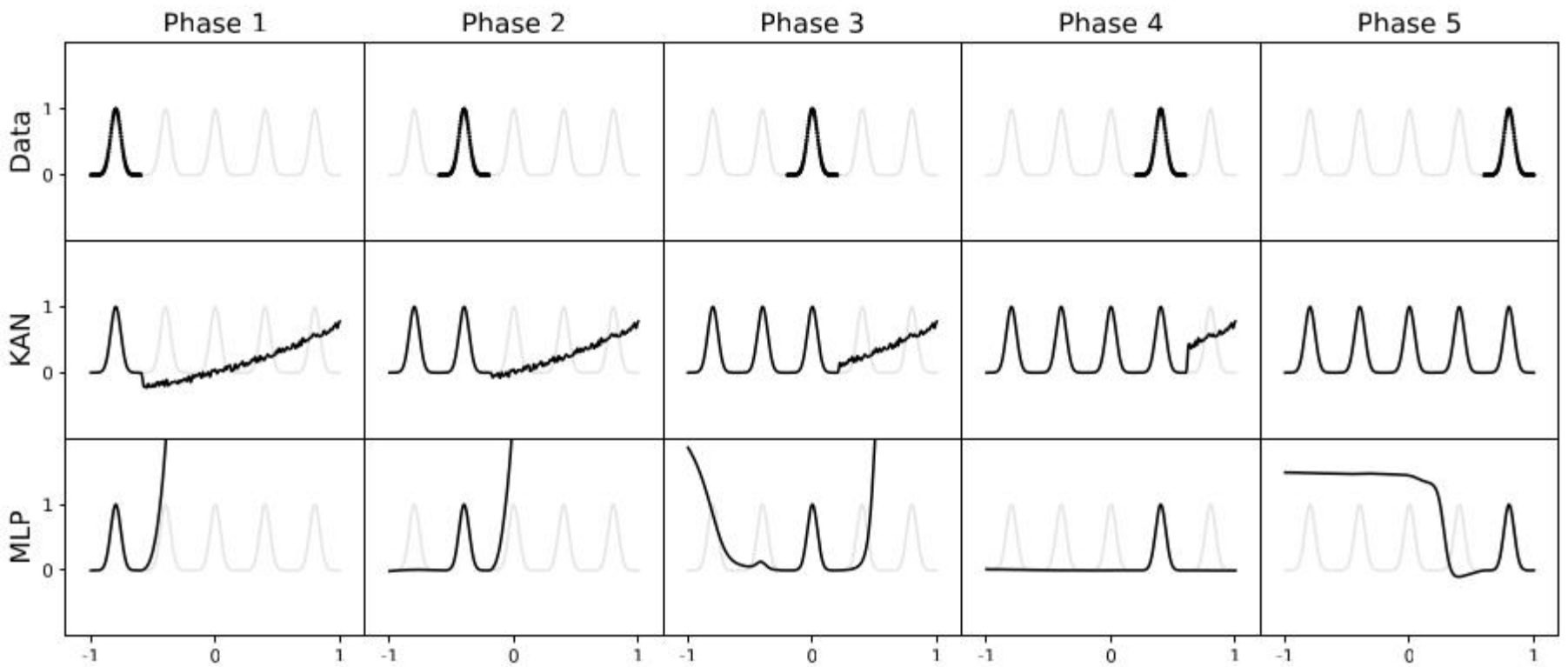


Interpretability & Discovery



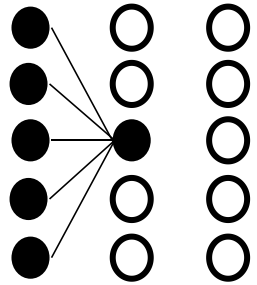
The MLP Assumption

Catastrophic Forgetting

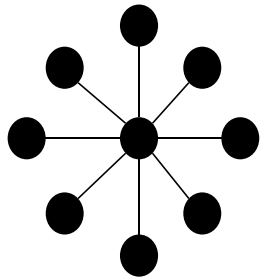


Inductive Bias & Self-Coordinating Networks

Dense and Feed-forward Style



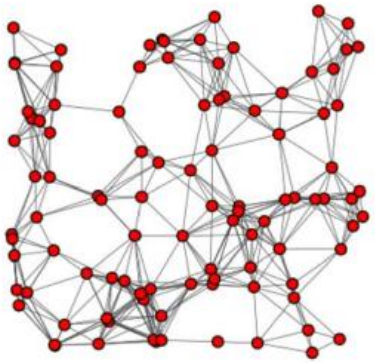
Self-Organizing Networks



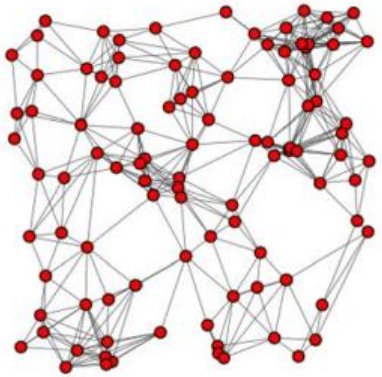
Biological & Physics Inspired Optimization

$$H_i = \alpha_i k_i^2 + \beta_i k_i^3 + \gamma_i r_i^2 + \lambda_i \sum_{j \neq i} \frac{A_{ij}}{d_{ij}}$$

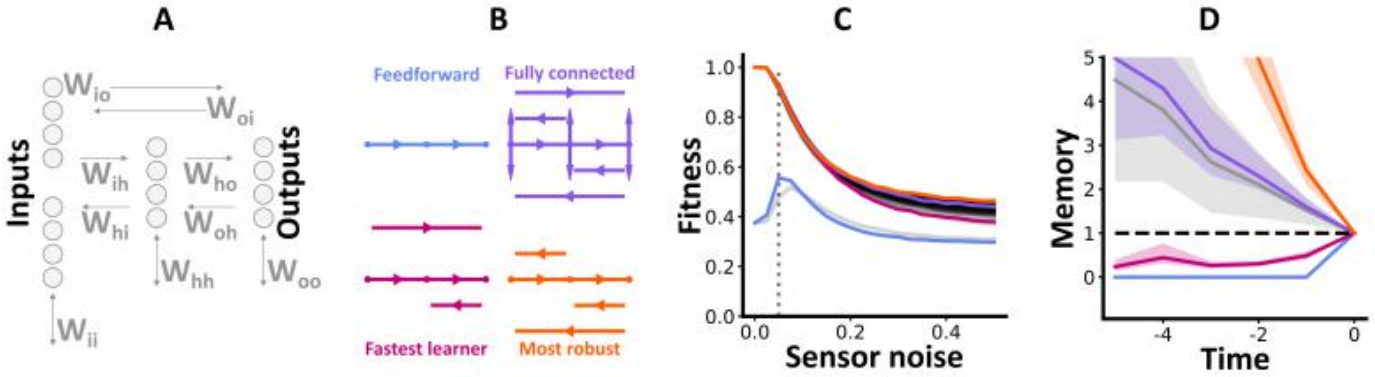
Static Node



Adaptive Node

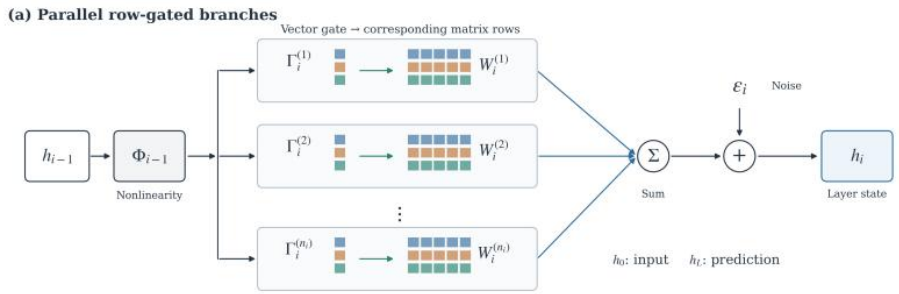


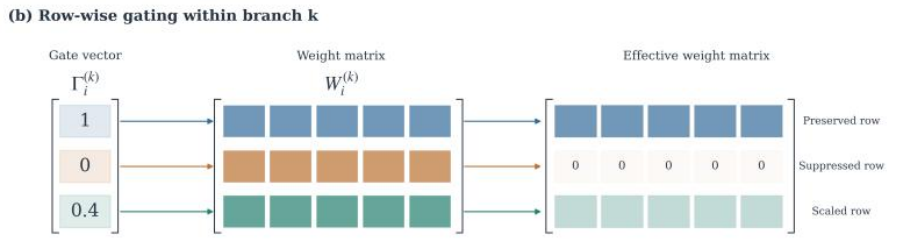
pRNNs

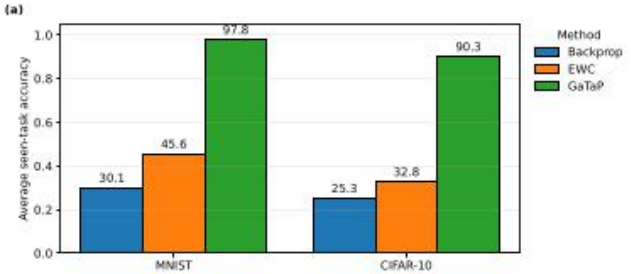


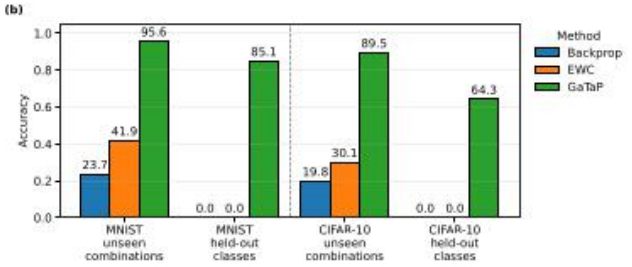
Inductive Bias & Self-Coordinating Networks

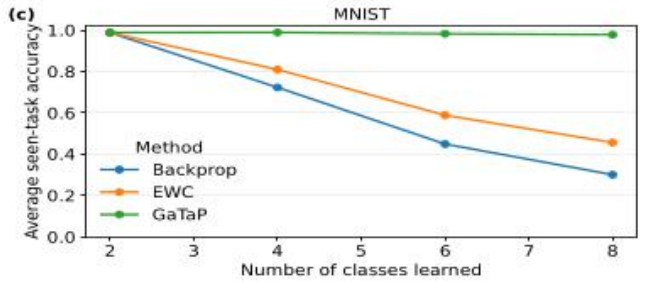
GaTaP

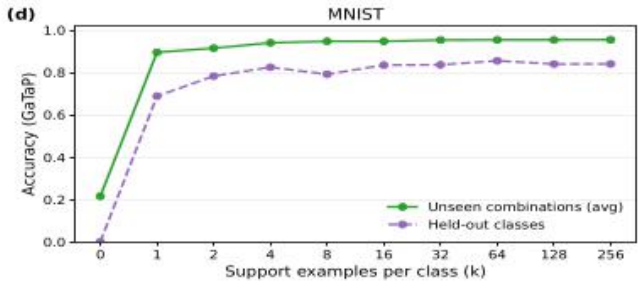




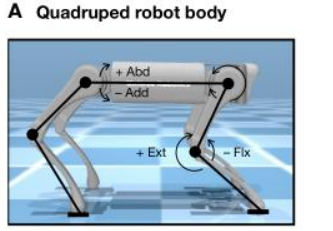






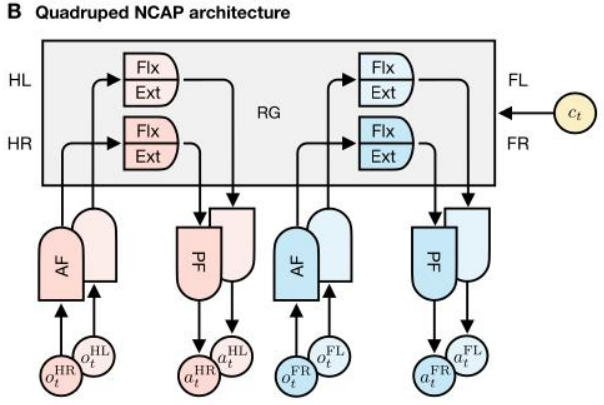


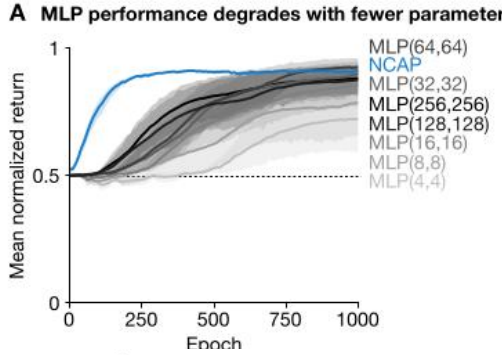
NCAP

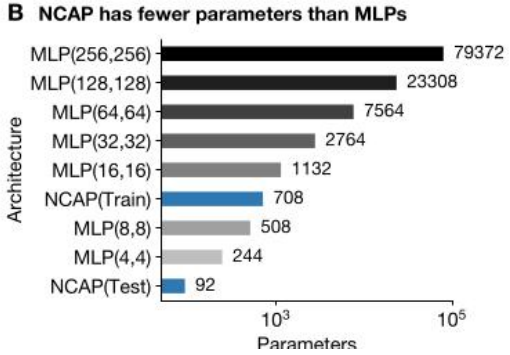


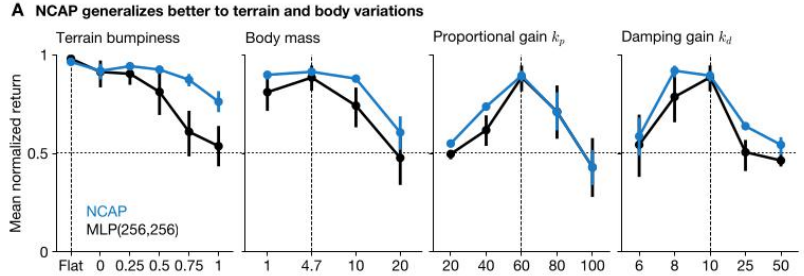
Observations: Joint positions, Joint velocities, Joint torques, Foot contact pressures

Actions: Joint position targets









$1Million Research

PHASE 1: PROOF OF CONCEPT (Months 0-18)

1. ***NCAP + GaTaP:***
   - *Show NCAP-CPG provide **natural gait** with **1000x parameter reduction***
   - *Show GaTaP enables **few/zero-shot terrain switching** with **zero catastrophic forgetting***
2. *Physical Hardware Validation:*
   - *Deply GaTaP-NCAP controllers on **Unitree A1 quadrupeds** and bipedal platforms*
   - *Demonstrate continual adaptation across changing environments*
   - *No policy retraining/simulator training*
3. *Ultra-Low Compute (On Edge Devices):*
   - *Run complex locomotion policies on **low-cost microcontroller/edge-chips***

PHASE 2: STRATEGIC FLYWHEEL (Months 18-24)

1. ***Improve Metrics:***
   - ***Improve sample efficiency by 10x***
   - ***Improve parameter count by 100x***
   - ***zero-shot sim-to-real transfer***
2. *Raise More Funds:*

PHASE 3: LONG TERM VISION (Years 2-5)

1. ***Scale Beyond Motor-Control:***
   - *Extend to self-coordinating, multi-agent swarms*
2. *Redundancy and Self-healing AI:*
   - *Research re-wiring and pruning of networks*
   - *Research adaptation to noise and hardware failures*

"The future of AI is not larger black boxes—it is biological parsimony. By uniting neural circuit priors (NCAP) with modular continual gating (GaTaP), we build physical robots that learn like animals: stable on day one, computationally frugal, and capable of infinite lifelong adaptation."

REFERENCES

1. *Bhattasali, Nikhil X., et al. "Neural circuit architectural priors for quadruped locomotion." arXiv preprint arXiv:2410.07174 (2024).*
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3. *Xu, Kunpeng, Lifei Chen, and Shengrui Wang. "Kolmogorov-arnold networks for time series: Bridging predictive power and interpretability." arXiv preprint arXiv:2406.02496 (2024).*
4. *Ghosh, Marcus, and Dan FM Goodman. "Partially recurrent neural networks maximize performance and minimize wiring."*
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Thank You !!!